

AOD: Adversarial Orthogonal Disentanglement for LVLM Hallucination Mitigation

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Figure 1: Seattle Mariners at Spring Training, 2010.

Abstract

While Large Vision-Language Models (LVLMs) have revolutionized multimodal understanding, object hallucination—generating descriptions conflicting with visual facts—remains a critical barrier to reliable deployment. Early hallucination mitigation approaches relying on external knowledge or instruction tuning incur high alignment tax and latency. Recent internal interventions also falter: attention-based sparsification creates a logical paradox by pruning via defective weights, while statistical classifiers on raw hidden states overlook the severe feature entanglement between truthful semantics and hallucinatory noise. To address these limitations, we propose Adversarial Orthogonal Disentanglement (AOD), which mitigates hallucinations via rigorous geometric feature separation of model hidden states. Moving beyond passive statistical probing, AOD orchestrates an end-to-end Min-Max adversarial game via gradient reversal layer. This mechanism mathematically enforces an orthogonal decomposition of hidden states into mutually exclusive truth and hallucination subspaces, theoretically guaranteeing maximized feature purity. Extensive experiments on various models and datasets demonstrate that AOD outperforms all baselines, effectively suppressing hallucinations while preserving semantic richness. Notably, the extracted hallucination component proves highly transferable, enabling robust mitigation across datasets.

CCS Concepts

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Keywords

Large Vision-Language Models, Hallucination Mitigation, Adversarial Disentanglement, Orthogonal Decomposition, Geometric Interpretation

ACM Reference Format:

Anonymous author. 2018. AOD: Adversarial Orthogonal Disentanglement for LVLM Hallucination Mitigation. In *Proceedings of Make sure to enter*

Unpublished working draft. Not for distribution.

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Conference acronym 'XX, Woodstock, NY

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ACM ISBN 978-1-4503-XXXX-X/2018/06

<https://doi.org/XXXXXXX.XXXXXXX>

2026-02-14 08:15. Page 1 of 1-4.

the correct conference title from your rights confirmation email (Conference acronym 'XX). ACM, New York, NY, USA, 4 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

While Large Vision-Language Models (LVLMs) demonstrate powerful multimodal capabilities [10], their tendency for object hallucination—generating content inconsistent with visual facts—presents a critical risk [11, 19]. This flaw compromises their reliability and safety, especially in high-stakes fields like autonomous driving and medical diagnosis [21]. Therefore, understanding the root mechanisms of these hallucinations and developing robust mitigation strategies is a paramount research priority.

Hallucination mitigation has progressed from external corrections to internal interventions, yet current methods suffer from fundamental limitations. External strategies like instruction tuning [4, 13] or retrieval-augmented generation (RAG) [9] incur prohibitive costs, either degrading reasoning through a high alignment tax or adding significant inference latency. Internal approaches are also flawed. Attention-based token pruning [15, 22] is paradoxical, as it uses the same defective attention weights that cause hallucinations, thereby reinforcing the error. Concurrently, research on hidden state intervention [5, 14] trains statistical classifiers that overlook the severe feature entanglement [17] between truthful and hallucinatory signals. This results in lossy interventions that suppress genuine details and produce impoverished text.

To address these limitations, we propose Adversarial Orthogonal Disentanglement (AOD), a framework shifting from statistical probing to geometric disentanglement for hallucination mitigation. AOD is based on the insight that hallucination is not random noise but a separable component residing in a subspace orthogonal to the non-hallucinatory subspace within the latent space. We first orchestrate a Min-Max adversarial game using a disentangler, a truth probe, and a hallucination adversary, governed by a gradient reversal layer [12]. Then we train the disentangler, aiming to maximize the truth probe's accuracy while also maximizing the hallucination adversary's entropy. Finally, this process enforces an orthogonal decomposition, creating a pure hallucination subspace that allows us to precisely remove hallucinatory noise without degrading truthful semantics. This disentangled hallucination component is highly transferable and can be directly subtracted from the initial tensors of other datasets to reduce hallucination.

Our main contributions are as follows:

- We propose AOD, a framework that introduces an adversarial game to enforce a geometric, orthogonal decomposition of the model's hidden state, rather than statistical classification.

- We isolate the hallucination component from the latent space, which allows for precise, lossless and plug-and-play intervention.
- Extensive experiments show AOD significantly outperforms all baselines. It not only suppresses hallucinations but also preserves semantic richness and demonstrates high transferability to unseen datasets.

2 Related Work

2.1 Hallucination Mitigation for LVLMS

External strategies for mitigating object hallucination, such as instruction tuning [8] or RAG [1], depend on outside knowledge. However, they incur a prohibitive alignment tax due to high computational costs and latency. Consequently, research has pivoted to internal interventions like attention-based sparsification [15, 20, 22]. By pruning tokens using attention scores, it trusts the same faulty mechanism that causes hallucinations, thereby risking the reinforcement of the initial error. Another stream intervenes in model hidden states. ASD [14] applies a linear shift based on the mean tensor difference, whereas TruthPrInt [5] trains non-linear classifier. Yet, by framing hallucination as classification, both methods overlook the entanglement of truthful and hallucinatory features, risking lossy interventions that erode fine-grained visual details.

2.2 Disentanglement for Hallucination Mitigation

Beyond its utility in hallucination quantification [2], disentanglement offers a principled framework for mitigation, isolating the underlying factors of variation that can lead to hallucinatory artifacts. One line of research employs causal intervention to distinguish genuine effects from spurious correlations within the data [6]. Another approach is to decompose features into discrete semantic components, such as separating class-specific attributes for few-shot learning [7] or disentangling spectral information in facial synthesis [18]. Recent work has shifted focus to the model's internal mechanisms, for instance by identifying and neutralizing disruptive attention hijackers that can otherwise corrupt the generation process [3].

3 Methodology

In this section, we present the Adversarial Orthogonal Disentanglement (AOD) framework as shown in Figure 2. Our approach is designed to isolate hallucination-related components from model hidden states through a joint optimization process, enabling both internal state purification and cross-dataset intervention.

3.1 Problem Formulation and Supervision Construction

We formalize the hallucination disentanglement task by analyzing the internal latent representations of an LVLMS $\mathcal{M}$. Given a dataset $\mathcal{D}$ containing m distinct samples, where each sample consists of an image I_i and a binary existence verification query q_i (e.g., "Does [object] exist in the image?") paired with a "Yes" or "No" response. For each sample, we extract the hidden activation vector $\mathbf{x}_i, l \in \mathbb{R}^d$ from the l -th transformer layer at the final token position

of the system prompt. Unlike methods that aggregate features across multiple depths, our framework constructs a layer-specific training set $\mathcal{D}_l = \{(\mathbf{x}_{i,l}, y_i)\}_{i=1}^m$. Here, the optimal layer index l is determined via ablation studies to identify the latent manifold where factual belief is most saliently encoded.

To supervise the disentanglement, we construct a binary factual consistency label y_i by measuring the alignment between the model's internal belief and external visual reality. The pre-generation belief $\hat{y}_i$ is derived from the output logits of the first generated token:

$$\hat{y}_i = \operatorname{argmax}_{w \in \{\text{yes}, \text{no}\}} P(w \mid I_i, o_i) \quad (1)$$

Let $g(I_i, o_i) \in \{0, 1\}$ denote the objective ground-truth existence of the queried object. The supervision signal is then defined as $y_i = \mathbb{I}[\hat{y}_i = g(I_i, o_i)]$, where $y_i = 1$ represents a truthful state and $y_i = 0$ identifies a hallucination event. This formulation maps the abstract concept of model consistency onto a concrete classification objective in the latent space.

3.2 Orthogonal Disentanglement and Adversarial Joint Optimization

Building upon the observation that hallucinatory content in LVLMS often exhibits linear separability [14], we propose an adaptive orthogonal decomposition to isolate these biased signals. Unlike current static mean-difference approaches that may inadvertently perturb essential task-relevant semantics, our framework seeks to identify a specific hallucination subspace while preserving the model's fundamental latent manifold through a geometric constraint.

We introduce a learnable unit vector $\mathbf{v} \in \mathbb{R}^d$ (where $\|\mathbf{v}\|_2 = 1$) to define the principal hallucination direction. For any hidden state $\mathbf{x} \in \mathcal{D}_l$, the decomposition is performed via a linear projection onto $\mathbf{v}$ and its orthogonal complement:

$$\mathbf{h}_{\text{hallu}} = (\mathbf{x}^\top \mathbf{v}) \cdot \mathbf{v}, \quad \mathbf{h}_{\text{truth}} = \mathbf{x} - \mathbf{h}_{\text{hallu}} \quad (2)$$

By defining $\mathbf{h}_{\text{truth}}$ as the residual, we impose a geometric orthogonality constraint ($\mathbf{h}_{\text{truth}} \perp \mathbf{h}_{\text{hallu}}$), which is intended to restrict the disentanglement process such that the hallucinatory components are isolated without altering the remaining representation along the truthful dimensions.

To achieve semantic purity within these subspaces, we formulate the disentanglement as a tripartite adversarial process involving the direction vector $\mathbf{v}$, a truth probe D_T (parameterized by θ_T), and a hallucination adversary D_A (parameterized by θ_A). We define the task loss $\mathcal{L}_{\text{task}}$ and the adversarial loss $\mathcal{L}_{\text{adv}}$ using binary cross-entropy (BCE) based on the consistency label y :

$$\mathcal{L}_{\text{task}}(\mathbf{v}, \theta_T) = -\mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}_l} [y \log D_T(\mathbf{h}_{\text{truth}}) + (1 - y) \log(1 - D_T(\mathbf{h}_{\text{truth}}))] \quad (3)$$

$$\mathcal{L}_{\text{adv}}(\mathbf{v}, \theta_A) = -\mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}_l} [y \log D_A(\mathbf{h}_{\text{hallu}}) + (1 - y) \log(1 - D_A(\mathbf{h}_{\text{hallu}}))] \quad (4)$$

where $D(\cdot) \in [0, 1]$ denotes the probability of the state being consistent ($y = 1$) predicted by the respective probes.

The core training logic is a saddle-point optimization problem where $\mathbf{v}$ is optimized to assist the truth probe while simultaneously

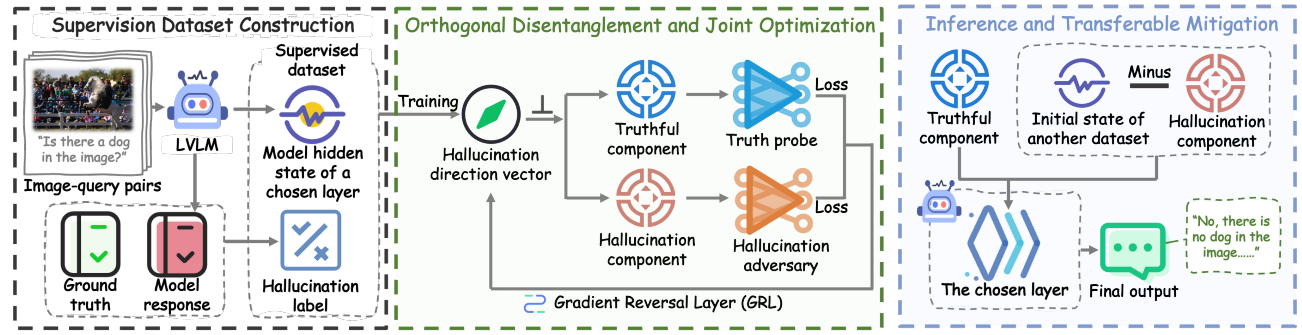


Figure 2: Pipeline of AOD. We first construct a labeled dataset by extracting hidden states from the LVLm and defining supervision signals based on the consistency between internal belief and visual facts. Then, we perform adversarial joint optimization to geometrically decompose the hidden state into orthogonal truthful and hallucinatory components, ensuring semantic disentanglement via GRL. Finally, during inference, we utilize the learned direction to subtract deceptive signals, enabling effective and transferable mitigation on unseen data.

deceiving the adversary. This objective is expressed as:

$$\min_{\mathbf{v}, \theta_T} \max_{\theta_A} (\mathcal{L}_{task}(\mathbf{v}, \theta_T) - \lambda \mathcal{L}_{adv}(\mathbf{v}, \theta_A)) \quad (5)$$

In this formulation, λ is a hyperparameter balancing the disentanglement strength. The inner maximization optimizes θ_A to extract label-relevant information from $\mathbf{h}_{hallu}$. In contrast, the outer minimization drives $\mathbf{v}$ to satisfy two competing requirements: minimizing $\mathcal{L}_{task}$ to ensure $\mathbf{h}_{truth}$ remains predictive, and maximizing $\mathcal{L}_{adv}$ to purge systematic bias from $\mathbf{h}_{hallu}$.

In practice, we implement this competitive process through an adversarial joint optimization utilizing a Gradient Reversal Layer (GRL) [16]. The GRL is placed between $\mathbf{h}_{hallu}$ and D_A , acting as an identity map during forward propagation but negating the gradients during backpropagation. Consequently, in a single backward pass, $\mathbf{v}$ receives gradients from two sources: a direct gradient from $\mathcal{L}_{task}$ and a reversed gradient from $\mathcal{L}_{adv}$. This co-evolution aims to reach a state where $\mathbf{h}_{truth}$ is maximally informative for consistency detection, while $\mathbf{h}_{hallu}$ is reduced to a semantic-free interference subspace.

3.3 Inference and Transferable Mitigation

The convergence of the AOD framework provides a dual-purpose mechanism for mitigating hallucinations during inference. On one hand, the purified truthful component $\mathbf{h}_{truth}$ can be utilized directly as a pristine representation for downstream reasoning, as it is stripped of the systematic interference that leads to internal belief mismatches. On the other hand, we discover that the isolated hallucination direction $\mathbf{v}$ captures a fundamental failure mode intrinsic to the model’s architecture rather than the training distribution. This suggests that $\mathbf{v}$ represents a universal hallucination manifold that exhibits significant transferability to unseen datasets.

Theoretical analysis implies that if $\mathbf{v}$ captures a cross-domain bias, the projection of a new hidden state $\mathbf{z}$ from an out-of-distribution dataset $\mathcal{D}'$ onto $\mathbf{v}$ should still isolate the hallucinatory interference specific to that state. We formalize this transferable mitigation as a latent steering operation:

$$\mathbf{z}' = \mathbf{z} - \alpha(\mathbf{z}^\top \mathbf{v})\mathbf{v} \quad (6)$$

where α is a scaling factor controlling the intervention strength. By subtracting the projection along $\mathbf{v}$, we effectively suppress the hallucinatory mode within $\mathbf{z}$ without requiring new labels or retraining. This geometric intervention demonstrates that the features identified by AOD are not merely overfitted patterns but are transferable latent structures, providing a lightweight yet robust tool for enhancing the factual consistency of LVLms across diverse and unseen visual-language tasks.

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Received 20 February 2007; revised 12 March 2009; accepted 5 June 2009